



Non-Equilibrium Thermodynamics: The Multispecies Boltzmann Equation

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Multispecies Boltzmann Equation

Vlasov–Boltzmann Equation



The *Vlasov–Boltzmann equation* for a single-species dilute gas is given as follows:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{a} \cdot \nabla_{\mathbf{v}} f = Q[f, f](\mathbf{x}, \mathbf{v}, t), \quad (1)$$

where f is the density of particles at position $\mathbf{x}$ with velocity $\mathbf{v}$ at time t ; and $\mathbf{a}$ is a particle acceleration determined by an external field, for example, an electric or magnetic field. The *Boltzmann transport equation* results from setting $\mathbf{a} \equiv \mathbf{0}$.

The Kinetic Density



The kinetic density $f = f(\mathbf{x}, \mathbf{v}, t)$ is a number density of atoms or particles. It should satisfy

$$f(\mathbf{x}, \mathbf{v}, t) \geq 0, \quad \forall \mathbf{x} \in \Omega, \quad \forall \mathbf{v} \in \mathbb{R}^d, \quad \forall t \geq 0,$$

and

$$\int_{\Omega} \int_{\mathbb{R}^d} f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v} d\mathbf{x} = N, \quad \forall t \geq 0,$$

where $N > 0$ is the number of particles in our physical box Ω .

Thus $p = p(\mathbf{x}, \mathbf{v}, t)$, defined by

$$p(\mathbf{x}, \mathbf{v}, t) := \frac{f(\mathbf{x}, \mathbf{v}, t)}{N},$$

is an evolving probability density.



A Really Simple Derivation

The derivation of the equation is a straightforward exercise using the chain rule. In particular, the total time derivative of the distribution f can be realized as

$$\begin{aligned}\frac{d}{dt}f(\mathbf{x}, \mathbf{v}, t) &= \sum_{i=1}^d \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} + \sum_{i=1}^d \frac{\partial f}{\partial v_i} \frac{dv_i}{dt} + \frac{\partial f}{\partial t} \\ &= \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{a} \cdot \nabla_{\mathbf{v}} f + \frac{\partial f}{\partial t},\end{aligned}$$

where $d = 3$ for the standard model.

Thus, the Vlasov–Boltzmann equation may be constructed via the following law: the total time rate of change of the distribution is equal to the collision (frequency) operator. In other words,

$$\frac{d}{dt}f(\mathbf{x}, \mathbf{v}, t) = Q[f, f](\mathbf{x}, \mathbf{v}, t).$$

The *Boltzmann collision operator*, $Q[f, f]$, requires much more physical insight for a clear derivation, involving certain conservation principles, in particular.



Boltzmann (Unary) Collision Operator

The Boltzmann collision operator is defined as

$$Q[f, f](\mathbf{x}, \mathbf{v}, t) = \frac{1}{m} \int_{\mathbb{R}^d} \int_{\mathbb{S}^{d-1}} [f' f'_* - f f_*] B(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) d\boldsymbol{\sigma} d\mathbf{v}_*,$$

where

$$\begin{aligned} f' &:= f(\mathbf{x}, \mathbf{v}', t), & f'_* &:= f(\mathbf{x}, \mathbf{v}'_*, t), \\ f &:= f(\mathbf{x}, \mathbf{v}, t), & f_* &:= f(\mathbf{x}, \mathbf{v}_*, t); \end{aligned}$$

m is the mass of an individual particle; B is the **collision kernel** describing binary collisions between particles; $\boldsymbol{\sigma}$ is a unit vector called the **apse direction**; and the primed coordinates are computed via the transformation

$$\mathbf{v}' = \mathbf{v} - \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*), \quad (2)$$

$$\mathbf{v}'_* = \mathbf{v}_* + \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*). \quad (3)$$



The BGK Approximation Model

BGK Approximation



The Boltzmann equation is usually too complicated to solve, even numerically, and is often approximated by replacing the collision operator.

The *Vlasov–BGK equation* is a model derived by approximating the Boltzmann collision operator with a simpler nonlinear, nonlocal relaxation operator of the form

$$Q^{BGK}[f](\mathbf{x}, \mathbf{v}, t) = \lambda(M_f(\mathbf{x}, \mathbf{v}, t) - f(\mathbf{x}, \mathbf{v}, t)), \quad (4)$$

where $\lambda = \frac{1}{\tau}$ is the collision frequency between particles, $\tau > 0$ is a characteristic time, and M_f is the **Maxwellian**.

The Maxwellian



The Maxwellian is constructed using the first three integral moments of f ,

$$n(\mathbf{x}, t) := \int_{\mathbb{R}^d} f \, d\mathbf{v}, \quad (5)$$

$$\mathbf{u}(\mathbf{x}, t) := \frac{1}{n} \int_{\mathbb{R}^d} f \mathbf{v} \, d\mathbf{v}, \quad (6)$$

$$\theta(\mathbf{x}, t) := \frac{m}{nd} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 \, d\mathbf{v}, \quad (7)$$

and is defined as

$$M_f(\mathbf{x}, \mathbf{v}, t) := M[f](\mathbf{x}, \mathbf{v}, t) := n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2\theta} \right). \quad (8)$$

The space-time fields n , $\mathbf{u}$, and θ , are the macroscopic (coarse grained) **number density**, **bulk velocity**, and **temperature**, respectively.

Other Macroscopic Variables



The following macroscopic variables are also of interest: the (kinetic) energy density is given by

$$\omega(\mathbf{x}, t) := \frac{m}{2} \int_{\mathbb{R}^d} f |\mathbf{v}|^2 d\mathbf{v}, \quad (9)$$

and the internal energy density is defined as

$$e(\mathbf{x}, t) := \frac{m}{2} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 d\mathbf{v}. \quad (10)$$

We observe that the temperature and the internal energy are related via

$$\theta(\mathbf{x}, t) = \frac{2}{d} \frac{e(\mathbf{x}, t)}{n(\mathbf{x}, t)}.$$



The Binary Collision Transformation



Multispecies Boltzmann Equation

Our goal is to make sense of the Multispecies Boltzmann equation. We define this as follows.

Let $\Omega \subset \mathbb{R}^3$ be a bounded open set. We say that the kinetic distributions $f_i : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$, $i = 1, \dots, R$, satisfy the **multi-species Boltzmann equation** iff, for all $\mathbf{x} \in \Omega$, $\mathbf{v} \in \mathbb{R}^3$, and $t \in (0, T]$,

$$\partial_t f_i(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i(\mathbf{x}, \mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t),$$

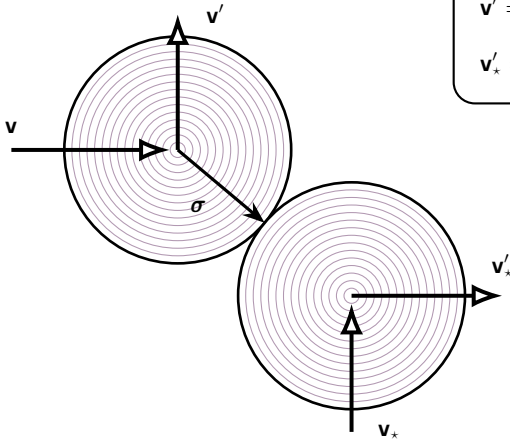
where $Q_i[f_i, \mathbf{f}]$, the **cumulative binary collision operator**, is defined as

$$Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) := \sum_{j=1}^R Q_{ij}[f_i, f_j](\mathbf{x}, \mathbf{v}, t).$$

We need a lot of machinery to understand what is going on here.



Binary Collisions



$$\mathbf{v}' = \mathbf{v} - \sigma [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)]$$

$$\mathbf{v}'_{\star} = \mathbf{v}_{\star} + \sigma [\sigma \cdot (\mathbf{v} - \mathbf{v}_{\star})]$$



Example

Suppose that, as in the figure on the previous slide, that the pre-collision velocities are

$$\mathbf{v} = [1, 0]^T, \quad \mathbf{v}_* = [0, 1]^T.$$

Assume the apse direction is

$$\boldsymbol{\sigma} = \frac{1}{\sqrt{2}}[1, -1]^T.$$

Then, it follows that, after the application of the collision transformation expressed on the previous slide (which assumes that the masses of the particles are the same), the post-collision velocities are

$$\mathbf{v}' = [0, 1]^T, \quad \mathbf{v}'_* = [1, 0]^T.$$



Example

Suppose that, as in the last example, that the pre-collision velocities are

$$\mathbf{v} = [1, 0]^T, \quad \mathbf{v}_* = [0, 1]^T.$$

But, now assume the apse direction is

$$\boldsymbol{\sigma} = [0, -1]^T.$$

Then, it follows that, after the application of the collision transformation, the post-collision velocities are

$$\mathbf{v}' = [1, 1]^T, \quad \mathbf{v}'_* = [0, 0]^T.$$

Does this make sense? Is this correct?



Definition (Binary Collision Transformation)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$. Fix a unit vector $\boldsymbol{\sigma} \in \mathbb{S}^2$. The transformation $T_{\boldsymbol{\sigma}}^{ij}(\mathbf{v}, \mathbf{v}_*) = (\mathbf{v}', \mathbf{v}'_*)$, which maps $\mathbb{R}^3 \times \mathbb{R}^3$ into $\mathbb{R}^3 \times \mathbb{R}^3$, is defined via

$$\mathbf{v}' = \mathbf{v} - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \boldsymbol{\sigma}, \quad (11)$$

$$\mathbf{v}'_* = \mathbf{v}_* + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma}. \quad (12)$$

We will write

$$\mathbf{v}' = T_{\boldsymbol{\sigma},1}^{ij}(\mathbf{v}, \mathbf{v}_*) \quad \text{and} \quad \mathbf{v}'_* = T_{\boldsymbol{\sigma},2}^{ij}(\mathbf{v}, \mathbf{v}_*)$$

for the first and second components of the transformation, respectively. The mapping $T_{\boldsymbol{\sigma}}^{ij}$ is called a **binary collision transformation** and $\boldsymbol{\sigma}$ is called the **apse direction**. The un-primed coordinates $\mathbf{v}$ and $\mathbf{v}_*$ are called the **pre-collision velocities** and the primed coordinates $\mathbf{v}'$ and $\mathbf{v}'_*$ are called the **post-collision velocities**.



Proposition (Conservation Properties)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$, and fix an apse direction $\sigma \in \mathbb{S}^2$. Suppose that $(\mathbf{v}', \mathbf{v}'_*) = T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)$. Then,

$$m_i \mathbf{v} + m_j \mathbf{v}_* = m_i \mathbf{v}' + m_j \mathbf{v}'_*, \quad (13)$$

$$\frac{m_i}{2} |\mathbf{v}|^2 + \frac{m_j}{2} |\mathbf{v}_*|^2 = \frac{m_i}{2} |\mathbf{v}'|^2 + \frac{m_j}{2} |\mathbf{v}'_*|^2. \quad (14)$$

Furthermore,

$$\sigma \cdot (\mathbf{v}' - \mathbf{v}'_*) = -\sigma \cdot (\mathbf{v} - \mathbf{v}_*) \quad (15)$$

and

$$|\mathbf{v}' - \mathbf{v}'_*| = |\mathbf{v} - \mathbf{v}_*|. \quad (16)$$



Proof.

To get (13) add m_i times (11) to m_j times (12), and the result immediately follows. To get (14), square both (11) and (12) to obtain

$$|\mathbf{v}'|^2 = |\mathbf{v}|^2 - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_j^2}{(m_i + m_j)^2},$$

$$|\mathbf{v}'_*|^2 = |\mathbf{v}_*|^2 + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_i}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v}_* + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i^2}{(m_i + m_j)^2}.$$



Proof. (Cont.)

Then,

$$\begin{aligned}
 m_i |\mathbf{v}'|^2 + m_j |\mathbf{v}'_\star|^2 &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_\star|^2 \\
 &\quad - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)] \frac{4m_i m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)] \frac{4m_i m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v}_\star \\
 &\quad + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)]^2 \frac{4m_i m_j^2}{(m_i + m_j)^2} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)]^2 \frac{4m_i^2 m_j}{(m_i + m_j)^2} \\
 &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_\star|^2 \\
 &\quad - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)]^2 \frac{4m_i m_j}{m_i + m_j} \\
 &\quad + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_\star)]^2 \frac{4m_i m_j}{(m_i + m_j)^2} (m_i + m_j) \\
 &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_\star|^2.
 \end{aligned}$$

The proofs of identities (15) and (16) are left as simple exercises. ■



Proposition (Involutory Property)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$, and fix an apse direction $\sigma \in \mathbb{S}^2$. Then, T_σ^{ij} is a linear transformation, that is,

$$T_\sigma^{ij}(\alpha(\mathbf{v}, \mathbf{v}_*) + \beta(\mathbf{w}, \mathbf{w}_*)) = \alpha T_\sigma^{ij}(\mathbf{v}, \mathbf{v}_*) + \beta T_\sigma^{ij}(\mathbf{w}, \mathbf{w}_*), \quad (17)$$

for all $\alpha, \beta \in \mathbb{R}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{v}_*, \mathbf{w}_* \in \mathbb{R}^3$. Furthermore,

$$(\mathbf{v}, \mathbf{v}_*) = T_\sigma^{ij}(T_\sigma^{ij}(\mathbf{v}, \mathbf{v}_*)), \quad (18)$$

for all $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$. In other words,

$$I = T_\sigma^{ij} \circ T_\sigma^{ij}, \quad (19)$$

or, equivalently,

$$\left(T_\sigma^{ij}\right)^{-1} = T_\sigma^{ij}. \quad (20)$$



Proof.

The linearity of T_{σ}^{ij} is a simple demonstration and is left to the reader. To prove the identity (18), one must show that

$$T_{\sigma,1}^{ij}(T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)) = \mathbf{v}$$

and

$$T_{\sigma,2}^{ij}(T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)) = \mathbf{v}_*.$$

These somewhat simple calculations are left to the reader. ■



Proposition (Transformation Symmetries)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$, and fix an apse direction $\sigma \in \mathbb{S}^2$. Then,

$$T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,2}^{j,i}(\mathbf{v}, \mathbf{v}_*), \quad (21)$$

$$T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,1}^{j,i}(\mathbf{v}, \mathbf{v}_*). \quad (22)$$

Proof.

Let $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$ be arbitrary. Then,

$$\begin{aligned} T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v}) &= \mathbf{v}_* - [\sigma \cdot (\mathbf{v}_* - \mathbf{v})] \frac{2m_j}{m_i + m_j} \sigma \\ &= \mathbf{v}_* + [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma \\ &= T_{\sigma,2}^{j,i}(\mathbf{v}, \mathbf{v}_*). \end{aligned}$$



Proof. (Cont.)

Similarly,

$$\begin{aligned}T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v}) &= \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v}_* - \mathbf{v})] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma} \\&= \mathbf{v} - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma} \\&= T_{\sigma,1}^{ji}(\mathbf{v}, \mathbf{v}_*).\end{aligned}$$





Proposition (Integration Domain Invariance)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$, and fix an apse direction $\sigma \in \mathbb{S}^2$. Suppose that $f : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ is integrable, that is,

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(\mathbf{v}, \mathbf{v}_*)| \, d\mathbf{v} d\mathbf{v}_* < \infty.$$

Then, $f \circ T_\sigma^{ij} : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ is integrable, and

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} f(\mathbf{v}, \mathbf{v}_*) \, d\mathbf{v} d\mathbf{v}_* = \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} f(T_\sigma^{ij}(\mathbf{v}, \mathbf{v}_*)) \, d\mathbf{v} d\mathbf{v}_*. \quad (23)$$

Proof.

This follows from the standard change of variable formula and the fact that $T_\sigma^{ij}(\mathbf{v}, \mathbf{v}_*)$ is a linear transformation with the property that

$$\left| \det \left(T_\sigma^{ij} \right) \right| = 1.$$





Definition (Binary Collision Operator)

Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, R\}$. Suppose that $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are integrable kinetic distributions. The **general binary i, j collision operator** is defined as

$$Q_{i,j}[f_i, f_j](\mathbf{v}) := \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) d\boldsymbol{\sigma} d\mathbf{v}_*, \quad (24)$$

where

$$\begin{aligned} f'_i &:= f_i(\mathbf{x}, \mathbf{v}', t), & f'_{j,*} &:= f_j(\mathbf{x}, \mathbf{v}'_*, t), \\ f_i &:= f_i(\mathbf{x}, \mathbf{v}, t), & f_{i,*} &:= f_i(\mathbf{x}, \mathbf{v}_*, t); \end{aligned}$$

and the primed coordinates are given precisely by the transformation

$$(\mathbf{v}', \mathbf{v}'_\star) = T_\sigma^{ij}(\mathbf{v}, \mathbf{v}_\star).$$

$B_{i,j} : \mathbb{R}^3 \times \mathbb{S}^2 \rightarrow [0, \infty)$ is the **collision kernel**.







Proposition (Prime Symmetry of the Collision Kernel)

Suppose that the collision kernel $B_{ij} : \mathbb{R}^3 \times \mathbb{S}^2 \rightarrow [0, \infty)$ has proper structure. Let $\mathbf{v}, \mathbf{v}_ \in \mathbb{R}^3$ be arbitrary pre-collisional velocities and set*

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*),$$

as usual. Then,

$$B_{ij}(\mathbf{v}' - \mathbf{v}'_*, \sigma) = B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma). \quad (27)$$



Proof.

Recall (15) and (16), respectively,

$$\boldsymbol{\sigma} \cdot (\mathbf{v}' - \mathbf{v}'_*) = -\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*) \quad \text{and} \quad |\mathbf{v}' - \mathbf{v}'_*| = |\mathbf{v} - \mathbf{v}_*|.$$

Then,

$$\begin{aligned} B_{ij}(\mathbf{v}' - \mathbf{v}'_*, \boldsymbol{\sigma}) &= b_{ij}(|(\mathbf{v}' - \mathbf{v}'_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v}' - \mathbf{v}'_*|) \\ &= b_{ij}(|-(\mathbf{v} - \mathbf{v}_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v} - \mathbf{v}_*|) \\ &= b_{ij}(|(\mathbf{v} - \mathbf{v}_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v} - \mathbf{v}_*|) \\ &= B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}), \end{aligned}$$

which establishes (27). ■



Definition (Multispecies Boltzmann Equation)

Let $\Omega \subset \mathbb{R}^3$ be a bounded open set. We say that the kinetic distributions $f_i : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$, $i = 1, \dots, r$, satisfy the **multi-species Boltzmann equation** iff, for all $\mathbf{x} \in \Omega$, $\mathbf{v} \in \mathbb{R}^3$, and $t \in (0, T]$,

$$\partial_t f_i(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i(\mathbf{x}, \mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t), \quad (28)$$

where $Q_i[f_i, \mathbf{f}]$, the **cumulative binary collision operator**, is defined as

$$Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) := \sum_{j=1}^R Q_{ij}[f_i, f_j](\mathbf{x}, \mathbf{v}, t). \quad (29)$$



Definition (Boundary Conditions)

We often assume that the physical boundary conditions are of the **specular reflection type**, that is, for all $(\mathbf{x}, \mathbf{v}) \in \Gamma_{\text{in}}$, where

$$\Gamma_{\text{in}} := \{(\mathbf{x}, \mathbf{v}) \in \partial\Omega \times \mathbb{R}^3 \mid \mathbf{v} \cdot \mathbf{n}(\mathbf{x}) < 0\},$$

and for all $t \in (0, T]$,

$$f_i(\mathbf{x}, \mathbf{v}, t) = f_i(\mathbf{x}, \mathbf{R}(\mathbf{x}, \mathbf{v}), t),$$

where

$$\mathbf{R}(\mathbf{x}, \mathbf{v}) := \mathbf{v} - 2[\mathbf{v} \cdot \mathbf{n}(\mathbf{x})]\mathbf{n}(\mathbf{x}),$$

and $\mathbf{n}(\mathbf{x})$ is the outward-pointing unit normal vector on the boundary at $\mathbf{x} \in \partial\Omega$.



Remark

In the following two results, we will assume that the kinetic distributions are functions only of velocity. This simplification reduces the number of independent variables that must be carried about. However, we lose no generality with these assumptions. More typically, the distributions are functions of space ($\mathbf{x}$) and time (t) as well.



Proposition (Collision Symmetry (Diagonal))

Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ and $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ are given and integrable. Then, assuming the collision kernel $B_{i,i}$ has proper structure,

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) \, d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi + \psi_* - \psi' - \psi'_*] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}, \end{aligned} \quad (30)$$

where the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,i}(\mathbf{v}, \mathbf{v}_*),$$

and, for shorthand

$$f_i := f_i(\mathbf{v}), \quad f_{i,*} := f_i(\mathbf{v}_*), \quad f'_i := f_i(\mathbf{v}'), \quad f'_{i,*} := f_i(\mathbf{v}'_*),$$

and

$$\psi := \psi(\mathbf{v}), \quad \psi_* := \psi(\mathbf{v}_*), \quad \psi' := \psi(\mathbf{v}'), \quad \psi'_* := \psi(\mathbf{v}'_*).$$



Proof.

Let us set

$$J = \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v}.$$

Then, by definition,

$$\begin{aligned} J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}. \end{aligned} \quad (31)$$



Proof. (Cont.)

The integration variables are dummy variables and may be interchanged. Doing so, we obtain

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{i,i}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\sigma,2}^{i,i}(\mathbf{v}_*, \mathbf{v})) B_{i,i}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v} d\mathbf{v}_* \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}_*) f_i(\mathbf{v}) B_{i,i}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v} d\mathbf{v}_* \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{i,i}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\sigma,2}^{i,i}(\mathbf{v}_*, \mathbf{v})) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v},
 \end{aligned}$$

where we have used (25).



Proof. (Cont.)

Observe that, using (21) and (22) (with $i = j$),

$$T_{\sigma,1}^{i,i}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*),$$

$$T_{\sigma,2}^{i,i}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*).$$



Proof. (Cont.)

Consequently,

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}_*) f_i(\mathbf{v}') B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v}. \tag{32}
 \end{aligned}$$



Proof. (Cont.)

Next, using the change of integration variables formula, and the fact that $l = T_{\sigma}^{i,i} \circ T_{\sigma}^{i,i}$, where l is the identity transformation, we have

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}'_*) B_{i,i}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}.
 \end{aligned}$$



Proof. (Cont.)

Using (27), we find

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &= -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}. \quad (33)
 \end{aligned}$$

Finally, with (32) as the starting point, using the change of integration variables formula, using the fact that $I = T_{\sigma}^{i,i} \circ T_{\sigma}^{i,i}$, and using the identity (27), we deduce that

$$J = -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}. \quad (34)$$

The combination of identities (31) – (34) gives the desired result. ■



Proposition (Collision Symmetry (Off Diagonal))

For $i \neq j$, if $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ and $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ are given and integrable, then, assuming the collision kernel $B_{i,j}$ has proper structure,

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi - \psi'] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\ &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi_* - \psi'_*] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \end{aligned} \quad (35)$$

where, to be precise, for the first integral the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,j}(\mathbf{v}, \mathbf{v}_*),$$

and, in shorthand,

$$\begin{aligned} f_i &:= f_i(\mathbf{v}), & f_{j,*} &:= f_j(\mathbf{v}_*), & f'_i &:= f_i(\mathbf{v}'), & f'_{j,*} &:= f_j(\mathbf{v}'_*), \\ \psi &:= \psi(\mathbf{v}), & \psi' &:= \psi(\mathbf{v}'). \end{aligned}$$



Proposition (Cont.)

For the second integral the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\sigma}^{j,i}(\mathbf{v}, \mathbf{v}_*),$$

and, in shorthand,

$$\begin{aligned} f_j &:= f_j(\mathbf{v}), & f_{i,*} &:= f_i(\mathbf{v}_*), & f'_j &:= f_j(\mathbf{v}'), & f'_{i,*} &:= f_i(\mathbf{v}'_*), \\ \psi_* &:= \psi(\mathbf{v}_*), & \psi'_* &:= \psi(\mathbf{v}'_*). \end{aligned}$$

Proof.

The steps are similar to those in the proof of the previous result. As before, let us set

$$J = \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) \, d\mathbf{v}.$$



Proof. (Cont.)

Then, by definition,

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{ij}(\mathbf{v}, \mathbf{v}_*)) f_j(T_{\boldsymbol{\sigma},2}^{ij}(\mathbf{v}, \mathbf{v}_*)) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{ij}). \quad (36)
 \end{aligned}$$



Proof. (Cont.)

Interchanging the integration variables $\mathbf{v}$ and $\mathbf{v}_*$, we obtain

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v})) f_j(T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v})) B_{ij}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) \, d\sigma \, d\mathbf{v} \, d\mathbf{v}_* \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}_*) f_j(\mathbf{v}) B_{ij}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) \, d\sigma \, d\mathbf{v} \, d\mathbf{v}_* \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v})) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) \, d\sigma \, d\mathbf{v}_* \, d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(\mathbf{v}) f_i(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) \, d\sigma \, d\mathbf{v}_* \, d\mathbf{v},
 \end{aligned}$$

where we have used (25).



Proof. (Cont.)

Observe that, using (21) and (22) (with $i \neq j$) and (26), we find that

$$\begin{aligned}
 J &= \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(T_{\sigma,1}^{j,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,2}^{j,i}(\mathbf{v}, \mathbf{v}_*)) B_{j,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(\mathbf{v}) f_i(\mathbf{v}_*) B_{j,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\sigma}^{j,i}).
 \end{aligned} \tag{37}$$



Proof. (Cont.)

Next, using the change of integration variables formula and the fact that $l = T_{\sigma}^{ij} \circ T_{\sigma}^{ij}$, where l is the identity transformation, we have

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{ij}(\mathbf{v}, \mathbf{v}_*)) f_j(T_{\sigma,2}^{ij}(\mathbf{v}, \mathbf{v}_*)) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_j(\mathbf{v}'_*) B_{ij}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}.
 \end{aligned}$$



Proof. (Cont.)

Using (27), we then find

$$\begin{aligned}
 J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_j(\mathbf{v}'_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\
 &= -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}') d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}).
 \end{aligned} \tag{38}$$

Finally, with (37) as the starting point, using the change of integration variables formula, using the fact that $I = T_{\boldsymbol{\sigma}}^{j,i} \circ T_{\boldsymbol{\sigma}}^{j,i}$, and using the identity (27), we deduce that

$$J = -\frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{ji}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}'_*) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}). \tag{39}$$

The combination of identities (36) – (39) gives the desired result. ■